

ARCHANA KANNAN

AI-Powered Software Quality Engineer

TypeScript | Playwright | WebdriverIO | Cypress | WEB/API/Mobile | Agentic AI | Vue.js | Azure DevOps

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EXECUTIVE PROFILE

- AI-POWERED SOFTWARE QUALITY ENGINEER - 16+ years across software testing, automation and end-to-end quality engineering, evolving traditional test automation into AI-powered quality engineering.
- PLAYWRIGHT, WEBDRIVERIO & CYPRESS AUTOMATION - 8+ years of hands-on automation experience, including Playwright using TypeScript/Node.js, WebdriverIO using JavaScript, and Cypress, across web UI, REST API, integration and E2E testing.
- AI-POWERED AUTOMATION - Modernises rule-based and traditionally hand-authored automation by using GitHub Copilot and Claude-based agents to interpret specifications, generate Playwright API tests and accelerate automation development.
- DEVELOPER-IN-TEST - Independently diagnoses and fixes defects in Vue.js front-end code, then validates changes through targeted regression testing.
- AUTOMATION ARCHITECT - Designs automation frameworks from scratch, defines reusable engineering standards and drives migration from Serenity BDD to Playwright with AI-assisted workflows.
- END-TO-END OWNERSHIP - Takes issues from defect discovery and root-cause analysis through functional/API validation and automated regression coverage.
- CI/CD & DEVOPS - Integrates smoke and regression automation into Azure DevOps pipelines and publishes execution outcomes through quality dashboards.
- INNOVATION & LEADERSHIP - Creates desktop automation tooling, AI-assisted engineering utilities and VS Code extensions; enables QA teams through training and technical leadership.

CAREER HIGHLIGHTS

- Led a Playwright/TypeScript proof of concept that secured management approval to modernise the automation stack from Serenity BDD to Playwright and trained the QA team on the new framework.
- Architected a reusable Playwright automation framework from scratch using TypeScript and Node.js for web UI, REST API, integration and end-to-end testing.
- Transformed traditional automation development toward AI-powered automation by building GitHub Copilot and Claude agents that consume API documentation/OpenAPI specifications and generate Playwright API tests.
- Debugs and fixes application defects directly in Vue.js code during testing, reducing QA-to-development hand-offs and accelerating defect resolution.
- Developed a reusable Playwright desktop automation package and a custom VS Code extension for Azure DevOps pull-request review workflows.
- Integrated smoke/regression suites into Azure DevOps CI/CD and onboarded execution outcomes to custom dashboards for stakeholder visibility.
- Recognised with the Entrepreneurial Award at Broadridge Analytics Solutions for innovative thinking and technical leadership.

CORE TECHNICAL SKILLS

AI-Powered Quality Engineering	GitHub Copilot, Claude, Agentic AI, AI test-generation agents, prompt-driven automation, OpenAPI-driven test authoring, AI-assisted PR review
Automation	Playwright (TypeScript/Node.js), WebdriverIO (JavaScript), Cypress, Serenity BDD (Java), Cucumber/Gherkin, Page Object Model
Development & Defect Fixing	Vue.js front-end debugging and bug fixes; root-cause analysis; fix validation
API & Services	Playwright API, Postman (REST/JSON), SoapUI (SOAP/XML), OpenAPI/Swagger
Testing	UI, API, E2E, Integration, Regression, Smoke, Cross-Browser, Desktop, System/SIT, Mobile Android/iOS
CI/CD & Source Control	Azure DevOps Pipelines, Azure Services, Jenkins, GitHub Actions, Git, GitHub, GitLab, PR/MR reviews
Languages & Build	TypeScript, JavaScript, Java, Node.js, NPM, Maven
Tools & Delivery	Visual Studio, VS Code, Jira, X-Ray, HP ALM, Confluence, Agile/Scrum, BDD, Waterfall, SDLC/STLC

AI-POWERED QUALITY ENGINEERING CAPABILITIES

- AGENTIC TEST AUTOMATION - Build AI agents that turn API specifications and documented test intent into reusable Playwright automation, reducing repetitive manual test-authoring effort.
- TRADITIONAL-TO-AI AUTOMATION MODERNISATION - Evolve established automation suites and engineering practices toward AI-assisted, specification-driven and reusable workflows while retaining deterministic validation and review.
- VUE.JS DEFECT RESOLUTION - Diagnose defects found through functional and automated testing, trace failures into Vue.js application code, implement focused fixes and validate behaviour end to end.
- AUTOMATION & CI/CD - Design reusable test architecture, integrate quality checks into deployment pipelines and maintain clear visibility of execution outcomes.
- AGILE QUALITY LEADERSHIP - Partner with Product Owners, SMEs and developers across refinement, planning, delivery, demonstrations and release readiness; coach colleagues on Playwright and AI-assisted automation adoption.

PROFESSIONAL EXPERIENCE

Senior QA Automation Engineer / SDET | Broadridge Analytics Solutions **Nov 2021 - Present**

Birmingham / Bristol, UK | Solvency II Manager - Finance / Asset Management reporting platform

- Led the Playwright/TypeScript proof of concept, demonstrated technical and delivery benefits to management, gained approval to migrate from Serenity BDD and supported adoption through team training.
- Designed and built the Playwright automation framework from scratch using TypeScript and Node.js, including architecture, folder structure, coding standards, reusable utilities and resilient locator strategies.
- Driving the evolution from traditional automation to AI-powered automation by creating agentic workflows that use GitHub Copilot and Claude to interpret API documentation/OpenAPI specifications and generate Playwright API automation tests.
- Automates REST/JSON APIs and web UI flows across end-to-end and integration scenarios for the Solvency II reporting platform.
- Independently debugs and fixes defects in Vue.js front-end code, performs root-cause analysis and validates changes with functional, API and Playwright regression tests.
- Developed a reusable Playwright desktop automation package and a VS Code extension integrated with Azure DevOps for automated PR review workflows.
- Integrated smoke/regression automation into Azure DevOps deployment pipelines and onboarded results to custom test harness dashboards.
- Maintained legacy Serenity BDD coverage during the Playwright migration and manages Git repositories, PR/MR workflows, peer reviews, merge conflicts, Jira defects and release-quality reporting.

QA Automation Engineer | Tata Consultancy Services (TCS) **Oct 2018 - Jun 2020**

Bristol, UK | Lloyd's Banking Group (Closed Book) - Maintenance releases

- Built and maintained WebdriverIO automation using JavaScript, Cucumber and Page Object Model.
- Automated functional and API regression scenarios for maintenance releases.
- Collaborated with business SMEs on test coverage and supported GitLab code reviews and merge requests.

QA Analyst / Project Lead | Tata Consultancy Services (TCS) **Aug 2010 - Sep 2018**

Chennai, India | Staples-Quill Retail E-Commerce & AXA General Insurance

- Led functional, integration and mobile testing for B2B/B2C e-commerce applications.
- Managed test planning, execution, defect tracking and release readiness activities.
- Performed functional testing for motor-insurance workflows including new business, renewals and claims.

QA Analyst | Cognizant Technology Solutions **May 2007 - Aug 2010**

Chennai, India | ACES Insurance - Dental Insurance, USA

- Performed functional testing for a US dental-insurance application across policy and claims workflows.
- Supported test design, execution, defect management and test reporting.

INDUSTRY, EDUCATION & RECOGNITION

Domains: Finance / Solvency II; Banking; General, Motor & Dental Insurance; Retail / B2B & B2C E-Commerce

Education: MBA - Anna University, Chennai, India | B.Sc. Mathematics - University of Madras, Chennai, India

Award: Entrepreneurial Award - Broadridge Analytics Solutions, recognised for innovative thinking and technical leadership